

Unequal Job Opportunities and Well-Being Inequality: A Latent-Jobs Structural Decomposition

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France, EU-SILC priced through EUROMOD.

Seminar of 17 September. Well-being is measured from two perspectives: the attained bundle (ATT) and the ex-ante opportunity prospect (EA).

Question and contribution

Two people with the same tastes and the same wage need not face the same set of jobs.

- How much of measured welfare inequality is attributable to unequal job access and earning opportunities, rather than to preferences?
- What does a model that assumes a common choice set do to that attribution?

Both questions are about a *measure*, not about a policy counterfactual.

The contribution is a structural opportunity set plus a welfare measure defined on it.

- A random-utility, random-opportunity (RURO) model in which *which jobs a household can reach* is estimated, not assumed.
- Two money-metric welfare perspectives: the bundle a household attains (ATT) and the opportunity prospect it faces (EA).
- An exact Shapley decomposition over preferences, geographic/temporal access and earning opportunities, holding household resources, needs and composition fixed.

The current decomposition equalises preferences, geographic/temporal job access and systematic earning opportunities while holding household resources, needs and composition fixed.

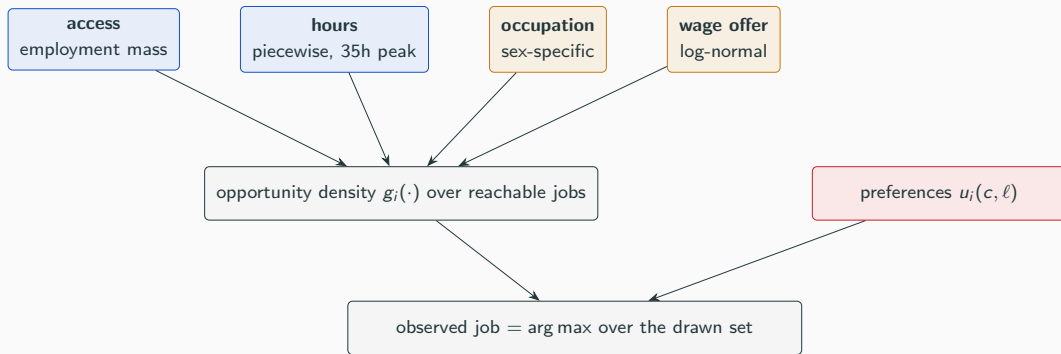
Data and the structural RURO model

France, EU-SILC priced through EUROMOD: 1 540 single-adult and 2 223 couple households.

- Estimation frames: final samples and the current estimated specification.
- Estimation frame: each household's observed job plus 100 drawn alternatives, every one priced through the tax–benefit system.
- Fit diagnostics use a separate, larger panel: 2 048 common quadrature nodes per household.
- Weights are the survey weights `dwt` throughout.

Results use a single specification throughout. Estimates from earlier, non-comparable sample frames are not shown.

A job is a package; a household draws from its own distribution over packages.



Access and capability are not separately identified; the access factor is their product.

Preferences and the opportunity density are estimated in one likelihood.

$$\Pr(j_i \mid x_i) \propto \exp\{ u_i(c_{ij}, \ell_{ij}) + \log g_i(j) - \log q_i(j) \}$$

u preferences g estimated opportunity density q numerical proposal density, divided out

The proposal correction is what makes the drawn alternatives a simulator for the estimated set rather than the choice set itself.

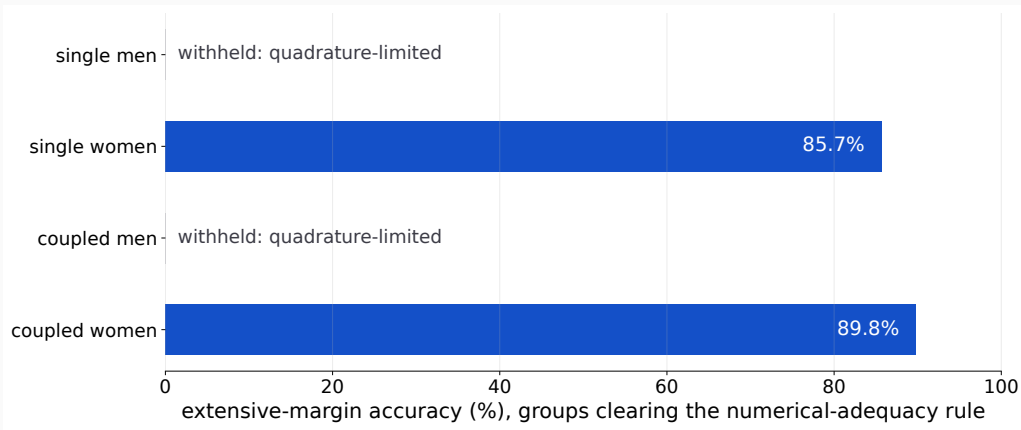
Positive estimates and fit

Estimated model: 41 free parameters for singles, 47 for couples.

	singles	couples
free parameters	41	47
estimation households	1 540	2 223
objective at $\hat{\theta}$	6253.463	10283.034

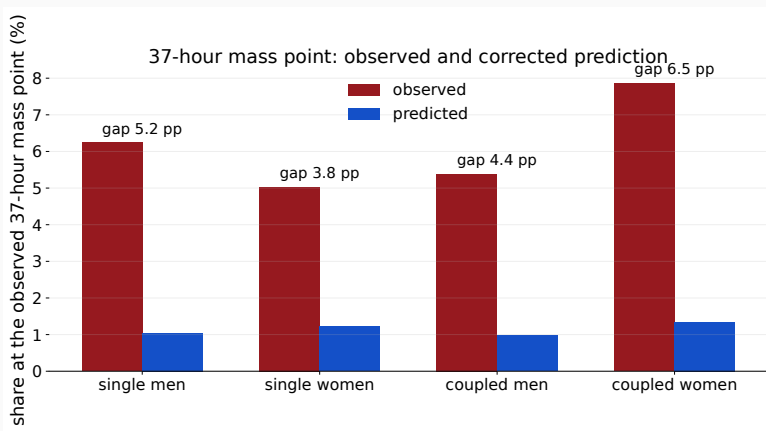
Objectives reproduce exactly under an independent diagnostics reimplementation.

Extensive accuracy is reportable for women in each sample; both groups of men are withheld.



The preregistered weighted numerical-adequacy threshold is unchanged. Single men and coupled men are quadrature-limited; their extensive accuracy is not reported.

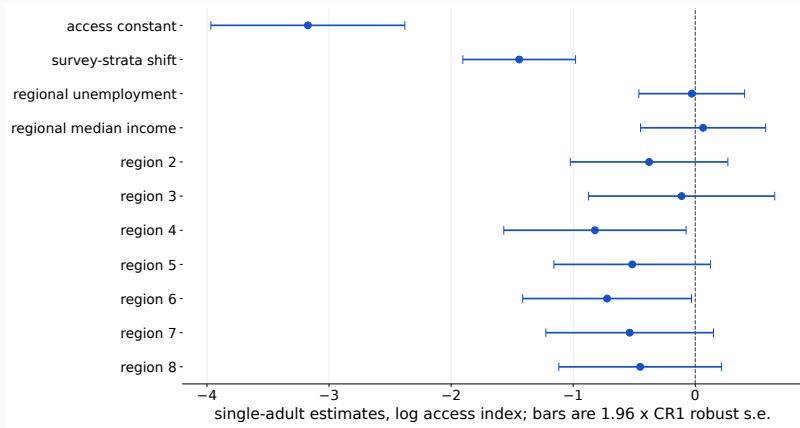
The remaining common hours mismatch is the observed 37-hour mass point, outside structural FT.



Underprediction of the observed 37-hour mass point is 3.8–6.5 percentage points across groups; this is not a structural full-time-band claim.

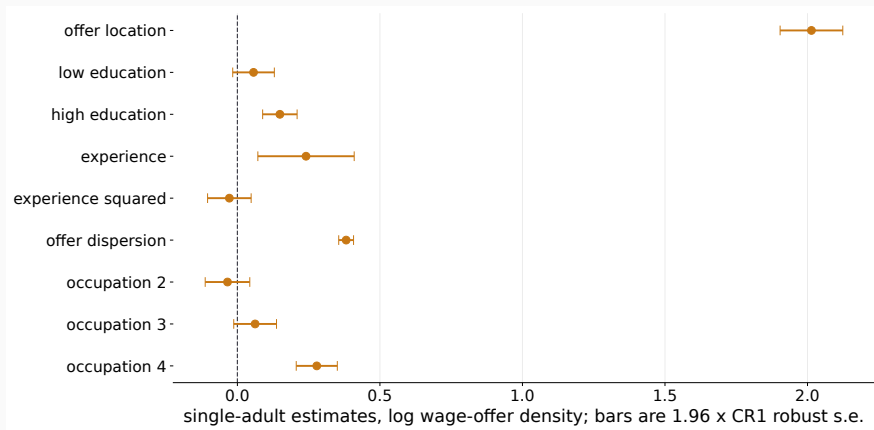
Evidence for heterogeneous opportunities

The access kernel is estimated, and it varies across households.



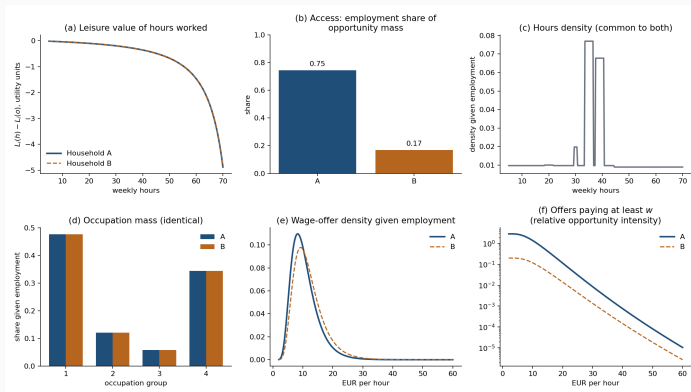
Single-adult estimates; horizontal bars are $1.96 \times \text{CR1 cluster-robust standard errors}$. Structural and model-conditional; not causal.

So is the earning-opportunity kernel: an offer distribution, not a single wage.



Single-adult estimates; occupation terms shift the offer location. Dispersion $\hat{\sigma} = 0.382$ ($z = 28.7$).

Two matched men, almost identical tastes, different opportunity environments.



Same occupation group, hours band and wage quintile. A has 14.5 times B's employment mass; B's wage-offer location is 11.5 log points higher. A stated-rule teaching example: not causal, not representative.

From choices to well-being: outcomes versus prospects

ATT: how well off is the household in the bundle it actually attains?

$$W_i^1 = m_i(o; z_i) \quad \text{where} \quad u_i(m_i(o), o) = u_i(z_i)$$

z_i attained bundle o the non-employment state m_i the money metric

The empirical measure uses the universally available non-employment state as its reference. A market-job reference is a sensitivity analysis that requires an unidentified intensity parameter; it is not in the baseline.

Under the current specification W^1 coincides with the staying-home equivalent — as a fact about this domain.

$$W_{i,F}^1 = C_i^{\text{obs}} \exp \left[\frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right]$$

- Inputs: C_i^{obs} , $L_i(j_i^{\text{obs}})$, $L_i(o)$, β_c — nothing else.
- No opportunity density g , no proposal q , no shock ε , no intensity κ .
- Current nonworkers: $W = C$; current workers: $W < C$ under the maintained empirical domain.

This is an **empirical-domain coincidence** under log consumption and universally available non-employment, not a general theoretical identity.

EA: how valuable is the distribution of job prospects the household faces?

$$J_i = \int e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} g_i(j) d\nu, \quad H_i = \int e^{L_i(j)} g_i(j) d\nu,$$

$$W_i^{EA} = \lambda_c \exp \left[\frac{\log J_i - \log H_i}{\beta_c} \right]$$

- J_i : the actual prospect over the whole estimated opportunity environment.
- H_i : same jobs and availability, equal consumption across jobs.
- W_i^{EA} : the flat consumption that makes the two prospects equally valuable.

The two measures answer different welfare questions. ATT evaluates the bundle eventually reached against a common non-employment reference; EA evaluates the whole distribution of potential job outcomes against a reference that keeps the opportunity environment and equalises consumption across jobs.

Attained-bundle distribution, equivalised

Baseline W^1 -F, equivalised, single adults.

	C^{eq}	$W_F^{1\text{eq}}$
households	1 540	1 540
weighted mean	1 766	1 302
weighted median	1 588	1 165
weighted Gini	0.263292	0.249807

Units: EUR/month, household-equivalised (modified-OECD scale); survey weight dwt. A descriptive dispersion comparison, not a welfare-loss index. The modified-OECD scale is used throughout the equivalised results. Singles and couples equivalised levels are never compared.

The sample aggregates reproduce to machine precision under an independent reimplementation. This is a descriptive dispersion comparison (C^{eq} against $W_F^{1\text{eq}}$ within the singles sample), not a welfare-loss index.

Baseline W^1 -F, equivalised, couples.

	C^{eq}	$W_F^{1\text{eq}}$
households	2 223	2 223
weighted mean	2 245	1 311
weighted median	2 063	1 237
weighted Gini	0.226805	0.197403

Units: EUR/month, household-equivalised (modified-OECD scale); survey weight dwt. A descriptive dispersion comparison, not a welfare-loss index. The modified-OECD scale is used throughout the equivalised results. Singles and couples equivalised levels are never compared.

Reported separately from singles: no pooled figure, and no cross-sample level comparison is drawn on this slide or elsewhere in this deck. The same independently reimplemented welfare construction and equivalence scale are used as for single adults.

The welfare question changes which labour-market inequality matters.

ATT: the attained bundle

	raw	equivalised
single adult	2.4%	1.9%
couple	6.7%	3.5%

Earning opportunities > access
in both populations and at both scales.

EA: the whole opportunity prospect

	raw	equivalised
single adult	14.8%	20.3%
couple	21.3%	7.9%

Singles: access $\simeq 3 \times$ earnings;
couples: no reversal, earnings > access.

The importance assigned to different labour-market inequalities depends on whether welfare evaluates the realised outcome or the opportunity prospect itself. Cells: access plus earning opportunities as a share of each perspective's own baseline Gini, holding resources, needs and composition fixed. Neither perspective is designated primary.

What I would like from you today.

- Is universally available non-employment the reference you would want for this question?
- Which counterfactual attainment estimand would you defend?
- What would convince you the access kernel is identified, given that access and capability are not separated?

Thank you.

Preliminary restricted-operator decomposition

The attained-bundle and ex-ante decompositions apply the same restricted three-pathway game while holding resources, needs and composition fixed. The ex-ante calculation is numerically certified; the attained-bundle counterfactual remains preliminary. Neither is a comprehensive or causal share of inequality due to all opportunities.

This bounded decomposition equalises three channels, not whole job access.

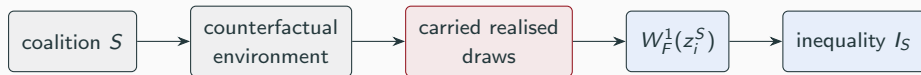
P systematic utility heterogeneity (tastes + reduced-form time constraints) $\rightarrow$ common reference

A local geographic/temporal access shifters (region, urban/rural, year) $\rightarrow$ common reference

B wage-offer location $\rightarrow$ common reference location

Household resources, needs and composition are held fixed; adding them as a fourth pathway D is a planned extension, and $A + B + D$ would not be an opportunity share. Personal occupation access and hours-band access are not equalised by A .

The bounded decomposition is executed; the final attainment estimand remains under design.



bounded decomposition route

This exercise reuses the already-priced estimation panel with no re-estimation and no new pricing. The final decomposition's counterfactual-attainment estimand remains under design.

Preliminary restricted-operator decomposition

In a preliminary three-factor structural exercise that holds household resources, needs and composition fixed, equalising systematic utility heterogeneity, coarse geographic/temporal access heterogeneity and earning opportunities changes money-metric well-being inequality by 1.8–9.9% of the baseline Gini, depending on household type and reporting scale. Within the Shapley allocation of that movable component, earning-opportunity heterogeneity has a larger contribution than the coarse geographic/temporal access channel in both samples and both reporting conventions. The preference contribution is not sign-robust to equalisation. This percentage is the change generated by the declared restricted P/A/B game. It is not an estimate of the total fraction of inequality caused by unequal job opportunities.

These are preliminary model-based accounting results, not causal estimates and not the final decomposition of total well-being inequality.

The earnings operator equalises wage-offer location only.

The earning-opportunity channel equalises wage-offer location only: the systematic differences associated with education and potential experience, which shift offer locations by at most about 0.10 log points. The common offer spread (about 0.37 log points), wage-draw luck and selection remain in the residual and are quantitatively larger than the location differences removed by this operator. This is a definitional boundary of the exercise, not a measurement error. Equalising locations can change attained wages; it does not equalise the whole wage distribution. The diagnostic comparison that also compresses the common spread and selection is a bound, not an additional decomposition result. Thin effective support for couples leaves individual wage attainment noisy and remains a numerical limitation.

Baseline W^1 -F, unequivalised, single adults (verification backup).

households	1 540
weighted mean	1 435
weighted median	1 320
weighted Gini	0.238
observed workers	1 336
observed non-workers	204

Units: household EUR/month, unequivalised; survey weight dwt. Descriptive baseline; no counterfactual, no decomposition, no causal reading.

Secondary/verification slide: the equivalised singles slide (§6) is primary. The sample aggregates reproduce to machine precision under an independent reimplementation.

Baseline W^1 -F, unequivalised, couples (verification backup).

households	2 223
weighted mean	2 496
weighted median	2 356
weighted Gini	0.210
observed workers	2 173
observed non-workers	50

Units: household EUR/month, unequivalised; survey weight dwt. Descriptive baseline; no counterfactual, no decomposition, no causal reading.

Secondary/verification slide: the equivalised couples slide (§6) is primary. Reported separately from singles. No pooled figure: the two samples are separate estimation frames and the welfare unit differs.

Pre-registered checks for the baseline W^1 -F measure

check	singles	couples
non-workers, $\max W^1 - C^{\text{obs}} $	0	0
workers, ratio outside $(0, 1)$	0	0
dependency test $(g, q, \varepsilon, \kappa)$	PASS	PASS
worked-household correspondence test	PASS	PASS
indifference identity, $\max \Delta u $	1.8e-15	1.8e-15
guards triggered	0	0

All six checks were pre-registered before execution. Worker-ratio violations were to be explained, never clipped; none occurred.

Quadrature adequacy for extensive-margin accuracy

group	extensive accuracy	status
coupled women	89.8%	reported (ratio 0.05)
single women	85.7%	reported (ratio 0.10)
coupled men	—	quadrature-limited, withheld
single men	—	quadrature-limited, withheld

Weighted, all households. Ratio is Monte Carlo s.e. over sampling s.d.; the pre-registered criterion is ≤ 0.25 . Node bootstrap over 2048 common quadrature nodes per household (diagnostic panel, not the estimation draws). Simulated bands: 500 outcome vectors at fixed $\hat{\theta}$. Source: individual-level predictive diagnostics.

Scope of the welfare evidence shown here

question	treatment here
empirical reference	home is selected on this empirical domain
alternative welfare construction	excluded from the baseline
measure correspondence	verified for worked households
counterfactual attainment	in design, not executed
reported scope	descriptive baseline and bounded preliminary decomposition
observed consumption	the estimated utility's own argument
alternative consumption benchmark	not constructed or reported
equivalised reporting	primary (§6), using the modified-OECD scale

Supporting technical provenance is retained in the source-only provenance block.